

Lecture 4: CNNs on Scientific Data
Blood Cells and Satellite Images
Finding T_c of the 2D Ising Model

Deep Learning for Scientific Discovery

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CNNs on Scientific Data: Blood Cells and Satellite Images



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MedMNIST

MNIST

Modified National Institute of Standards and Technology database: 70 000 grey images, 28×28 pixels, of hand-written digits $0 \ldots 9$ — our "Hello world".

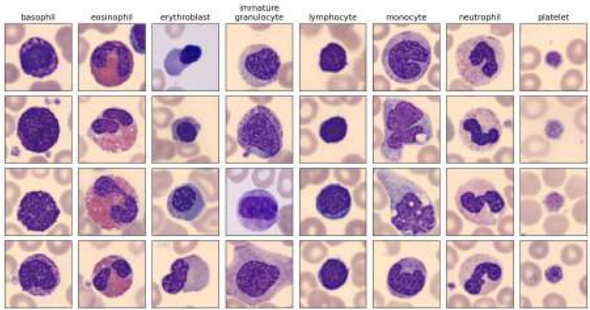
MedMNIST

the same idea for biomedical images: small images of a fixed size, a fixed split into training, validation and test part, one file per dataset.

- MedMNIST v2: 12 datasets of 2D images and 6 datasets of 3D images
- pathology slides, chest X-rays, skin lesions, retina scans, **blood cells**, organs in CT, ...
- images at 28×28 pixels; larger versions at 64, 128 and 224 pixels
- one line of code to load: `pip install medmnist`

J. Yang et al., *MedMNIST v2 — A large-scale lightweight benchmark for 2D and 3D biomedical image classification*, Sci. Data **10**, 41 (2023), doi:10.1038/s41597-022-01721-8; medmnist.com

BloodMNIST: Blood Cells under the Microscope



- 17 092 microscope images of single blood cells
- colour images, 64×64 pixels
- label: one of 8 cell types, assigned by clinical pathologists

The task:
image of one cell $\rightarrow$ its type, e.g. **"neutrophil"**

A. Acevedo et al., *A dataset of microscopic peripheral blood cell images for development of automatic recognition systems*, Data in Brief **30**, 105474 (2020), doi:10.1016/j.dib.2020.105474

The Eight Cell Types

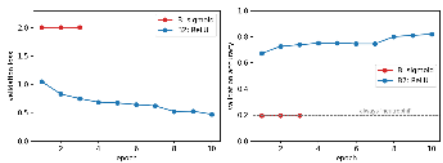
label		cell type	what it is	images
0		basophil	granulocyte with large dark granules	852
1		eosinophil	granulocyte with a two-lobed nucleus	2 181
2		erythroblast	immature red blood cell with a nucleus	1 085
3		immature granulocyte	precursor of granulocytes	2 026
4		lymphocyte	small cell with a large round nucleus	849
5		monocyte	largest white blood cell, folded nucleus	993
6		neutrophil	most common white blood cell, lobed nucleus	2 330
7		platelet	small cell fragment without a nucleus	1 643

The classes are **not balanced**: always answering "neutrophil" is right for 19.5% of the training images.

Blood Cells: the Models and Their Accuracy

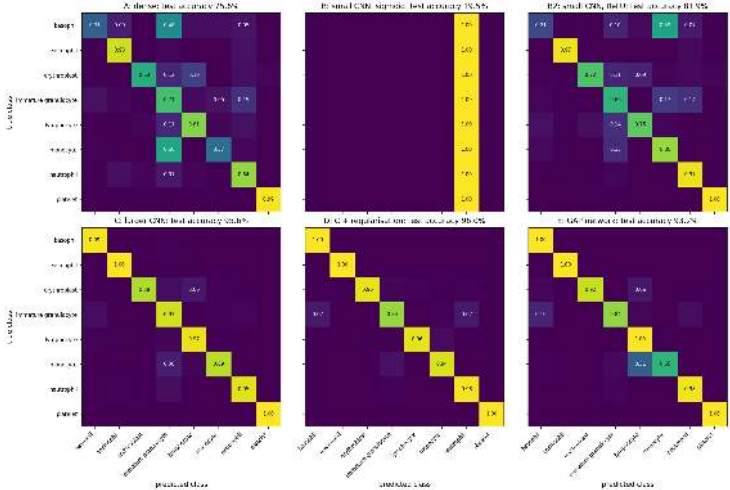
	model	parameters	train	val.	test
A	dense network on the flattened image	258 245	78.3%	75.2%	75.6%
B	small CNN, sigmoid	689 221	19.5%	19.5%	**19.5%**
B2	the same small CNN, ReLU	689 221	86.0%	82.0%	81.9%
C	larger CNN	4 292 168	99.2%	95.5%	95.6%
D	C + batch norm, dropout, L2, augmentation	4 292 616	96.2%	96.4%	96.0%
E	GAP network	288 936	93.5%	94.0%	93.2%

same data, loss and optimiser; 10 epochs each



- Model B answers "neutrophil" for every image: 19.5%, the share of neutrophils. Its loss, 2.003, is the entropy of the class frequencies.
- its sigmoids are saturated: the spread of their inputs grew from 0.35 to 98.60, and the gradient vanishes
- the same network with ReLU: **81.9%**

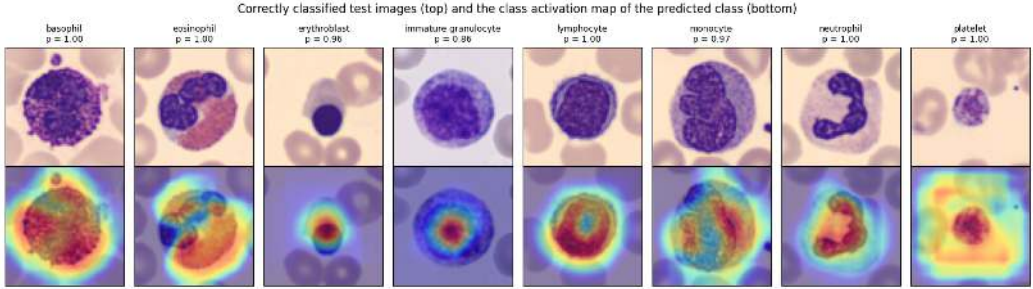
Blood Cells: Confusion Matrices of All Models



rows: true class; columns: predicted class

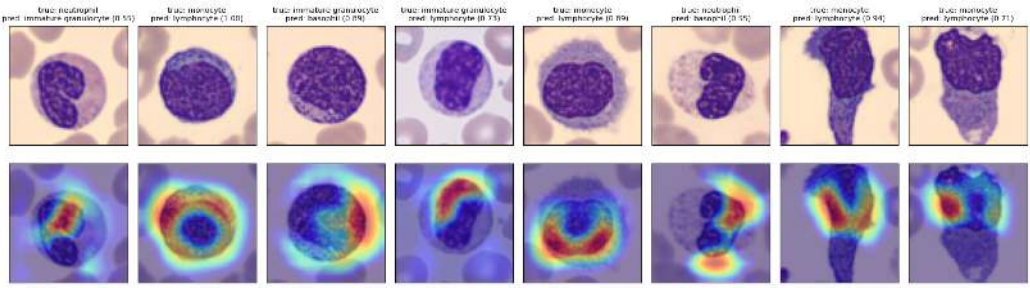
- A: 60% of monocytes taken for immature granulocytes
- B: one column, “neutrophil” for every image
- B2: 59% of basophils taken for monocytes
- C, D: almost diagonal
- E: 31% of monocytes taken for lymphocytes

CAM on Blood Cells



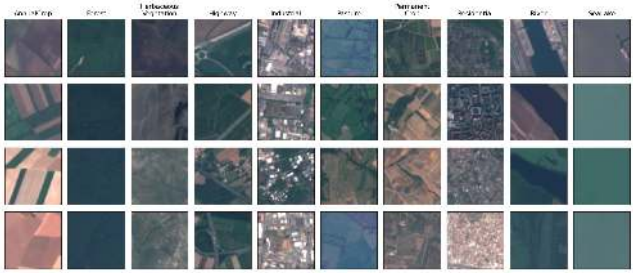
- top row: correctly classified test images, one per class
- bottom row: class activation map of the predicted class
- the evidence lies on the cell itself
- not on the red blood cells around it

Blood Cells: the Mistakes



- 233 of 3 421 test images are misclassified
- eight of them shown, with the map of the *predicted* class
- largest confusion: 31% of the monocytes are taken for lymphocytes
- an earlier run confused a different pair: errors are not stable from run to run

EuroSAT: Satellite Images of the Earth's Surface



- 27 000 patches from the Sentinel-2 satellites
- 64×64 pixels: 640 m $\times$ 640 m on the ground
- label: one of 10 land-use classes

The task:
one patch $\rightarrow$ its land-use class, e.g. **"River"**

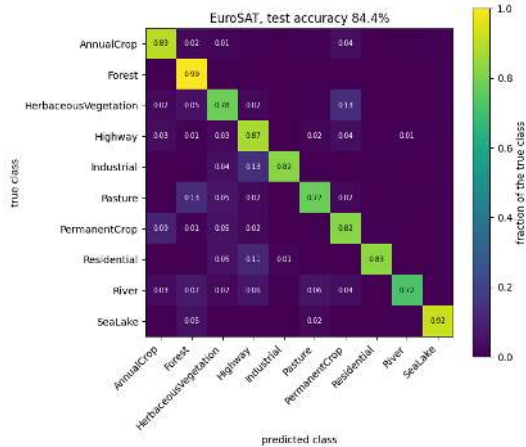
P. Helber et al., *EuroSAT: A Novel Dataset and Deep Learning Benchmark for Land Use and Land Cover Classification*, IEEE JSTARS **12**, 2217 (2019), doi:10.1109/JSTARS.2019.2918242

EuroSAT: Ten Land-Use Classes

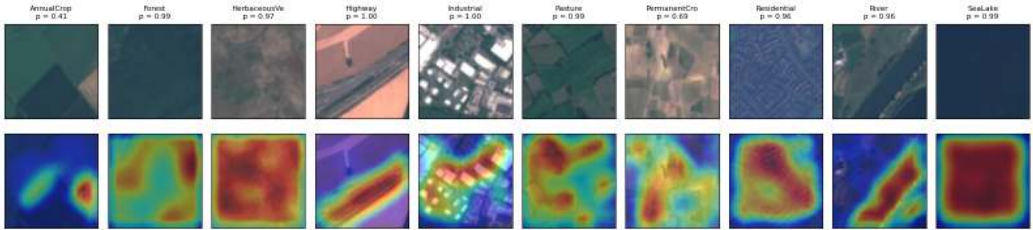
	class	on the ground	images
0	AnnualCrop	fields sown and harvested every year: wheat, maize	3 000
1	Forest	forest	3 000
2	HerbaceousVegetation	grass and other low natural vegetation	3 000
3	Highway	motorway or major road	2 500
4	Industrial	industrial buildings	2 500
5	Pasture	grassland for grazing animals	2 000
6	PermanentCrop	orchards, vineyards	2 500
7	Residential	residential buildings	3 000
8	River	a river crossing the patch	2 500
9	SeaLake	sea or lake	3 000

EuroSAT: Results

- the GAP network of the blood cells, with 10 outputs
- augmentation: flips, shifts, rotations by 90°
- 1 000 patches per class, 10 epochs: **84.4% test accuracy**
- best: Forest (99.5%); worst: River (72.2%)
- an earlier run of the same code: 90.0%



CAM on EuroSAT



- top row: one correctly classified test patch per class
- bottom row: class activation map of the predicted class
- highway, river: the map follows the object
- forest, sea: the map covers the whole patch

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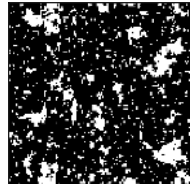
A Physics Example: the 2D Ising Model



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The Ising Model as a Grid of Cells

- a grid of cells, each in one of **two states** (spin up or down)
- each cell **prefers the state of its neighbours**
- **temperature** is random noise: it flips cells whatever their neighbours do
- low noise: one state spreads over the whole grid; high noise: a random mosaic. In between lies a sharp threshold, the **critical temperature** T_c



a sampled configuration at T_c: patches of all sizes

The same mathematics elsewhere

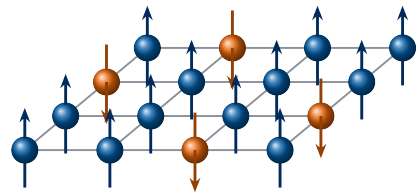
- **magnets:** iron is magnetic below its Curie temperature, $1043\,\mathrm{K}$ ($770\,^{\circ}\mathrm{C}$), and not above it
- **neurons:** the joint firing patterns of retinal neurons (active or silent) are described by an Ising model fitted to recordings
- **images:** neighbouring pixels of a map or a satellite image tend to share a class; Ising-type models are used as priors to restore and segment such images

Our question: can a network find T_c only by looking at snapshots of the grid, as it classified blood cells and land use?

E. Schneidman, M. J. Berry, R. Segev, W. Bialek, *Weak pairwise correlations imply strongly correlated network states in a neural population*, Nature **440**, 1007 (2006), doi:10.1038/nature04701; S. Geman, D. Geman, *Stochastic Relaxation, Gibbs Distributions, and the Bayesian Restoration of Images*, IEEE TPAMI **6**, 721 (1984), doi:10.1109/TPAMI.1984.4767596

The 2D Ising Model

- a square lattice $L \times L$; on every site i a *spin* $s_i = +1$ (up) or $s_i = -1$ (down)
- a simple model of a magnet: each spin is a tiny magnetic moment
- $2^{L \cdot L}$ possible configurations $\{s_i\}$



Energy of a configuration (Hamiltonian)

$$H(\{s_i\}) = -J \sum_{\langle i,j \rangle} s_i s_j - h \sum_i s_i$$

$\langle i, j \rangle$: pairs of nearest neighbours; $J > 0$: aligned neighbours lower the energy (ferromagnet), we set $J = 1$; h: external magnetic field, we set $h = 0$.

The Couplings Decide the Lowest-Energy State

$H = -J \sum_{\langle i,j \rangle} s_i s_j - h \sum_i s_i$: the sign of J decides which neighbours are favourable.
Lowest-energy configurations at $h = 0$:



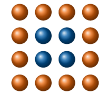
$J > 0$: ferromagnetic

neighbours *aligned*
all up or all down: 2 ground states
energy $-2J$ per spin
magnetisation $M = 1$



$J < 0$: antiferromagnetic

neighbours *opposite*
checkerboard pattern: 2 ground states
energy $-2|J|$ per spin
magnetisation $M = 0$



$J = 0$: non-interacting

spins independent
all $2^{L \cdot L}$ configurations have the same energy
no order at any temperature

- square lattice: each spin has 4 neighbours, so 2 bonds per spin
- $h = 0$: flipping *all* spins leaves H unchanged — hence two equivalent ground states
- $h > 0$: spins prefer $+1$; $h < 0$: spins prefer -1; the field removes this symmetry and selects one state

blue: $s_i = +1$, orange: $s_i = -1$. From here on: $J = 1$ (ferromagnet), $h = 0$.

The Partition Function

At temperature T ($k_B = 1$) every configuration appears with the Boltzmann probability

$$p(\{s_i\}) = \frac{e^{-H(\{s_i\})/T}}{Z}, \qquad Z(T) = \sum_{\{s_i\}} e^{-H(\{s_i\})/T}$$

Z, the **partition function**, is the normalisation: a sum over all $2^{L \cdot L}$ configurations.

Knowing Z gives the physical properties as functions of T ($\beta = 1/T$)		
mean energy	$\langle H \rangle = -\dfrac{\partial \ln Z}{\partial \beta}$	average energy of the configurations at T
specific heat	$C = \dfrac{d\langle H \rangle}{dT} = \beta^2 \dfrac{\partial^2 \ln Z}{\partial \beta^2}$	how fast the energy grows with T; proportional to the energy fluctuations
mean magnetisation	$\langle S \rangle = \dfrac{1}{\beta} \dfrac{\partial \ln Z}{\partial h}$, $S = \sum_i s_i$	net alignment of the spins: $\neq 0$ ordered, 0 disordered
susceptibility	$\chi = \dfrac{d\langle S \rangle}{dh} = \dfrac{1}{\beta} \dfrac{\partial^2 \ln Z}{\partial h^2}$	how strongly the spins follow a small field h; proportional to the magnetisation fluctuations

Solving the model means finding Z, or $\ln Z$, in analytical form as a function of (J, h, T); all equilibrium properties then follow by differentiation.

Onsager's Solution of the Partition Function

Every observable is a derivative of $\ln Z$, but $Z = \sum_{\{s_i\}} e^{-\beta H(\{s_i\})}$ is a sum over 2^N ($N = L^2$ spins) configurations. **Computing $\ln Z$ in closed form solves the model.**

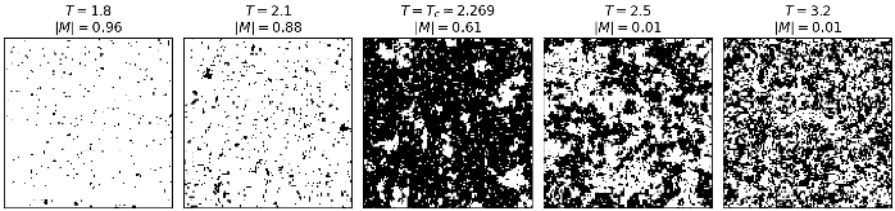
Onsager (1944): $\ln Z$ computed exactly for the 2D square lattice, $h = 0$, $N \to \infty$

$$\frac{1}{N}\ln Z = \ln\!\left(2\cosh(2\beta J)\right) + \frac{1}{2\pi}\int_0^{\pi} d\phi\, \ln\!\left[\tfrac{1}{2}\left(1+\sqrt{1-\kappa^2\sin^2\phi}\right)\right], \qquad \kappa = \frac{2\sinh(2\beta J)}{\cosh^2(2\beta J)}$$

- **method:** write $Z = \mathrm{Tr}\, V^L$, with V the $2^L \times 2^L$ *transfer matrix* that adds one row of spins; for a large lattice $\ln Z$ is fixed by the largest eigenvalue of V, and Onsager found all eigenvalues exactly
- $\kappa = 1$ exactly when $\sinh(2J/T) = 1$: there $\ln Z$ is **not analytic**, so $T_c = \dfrac{2J}{\ln(1+\sqrt{2})} \approx 2.269\, J$ (located already in 1941 by Kramers and Wannier from a low/high-temperature duality)
- near T_c: $C \sim -\ln|T - T_c|$, $\langle M\rangle \sim (T_c - T)^{1/8}$, $\chi \sim |T - T_c|^{-7/4}$, $\xi \sim |T - T_c|^{-1}$: a continuous transition with exponents $\alpha = 0$, $\beta = \frac{1}{8}$, $\gamma = \frac{7}{4}$, $\nu = 1$
- $\langle M\rangle = [1 - \sinh^{-4}(2J/T)]^{1/8}$ was announced by Onsager in 1949 and derived by C. N. Yang in 1952

H. A. Kramers, G. H. Wannier, *Statistics of the Two-Dimensional Ferromagnet. Part I*, Phys. Rev. **60**, 252 (1941), doi:10.1103/PhysRev.60.252; L. Onsager, *Crystal Statistics. I. A Two-Dimensional Model with an Order-Disorder Transition*, Phys. Rev. **65**, 117 (1944), doi:10.1103/PhysRev.65.117; C. N. Yang, *The Spontaneous Magnetization of a Two-Dimensional Ising Model*, Phys. Rev. **85**, 808 (1952), doi:10.1103/PhysRev.85.808

The Phase Transition in Pictures



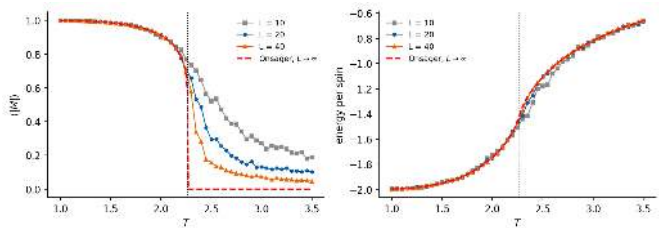
one sampled configuration at each temperature, 128×128 lattice ($N = 128^2$ spins); white and black: $s_i = +1$ and $s_i = -1$; $|M| = |\frac{1}{N}\sum_i s_i|$

- **below T_c:** the coupling between neighbours wins over thermal noise; almost all spins point the same way, up or down, and the lattice is a magnet by itself
- **above T_c:** thermal noise wins; small patches point randomly, $M = 0$
- **at T_c:** clusters of all sizes, from single spins to the size of the lattice: *critical behaviour*

In an infinite lattice M drops to zero exactly at T_c: a sharp change of the whole system caused by a small change of temperature.

A Phase Transition at a Critical Temperature

- **1D** (Ising, 1925): no phase transition at any $T > 0$
- **2D**: a phase transition at $T_c \approx 2.269$
- $T < T_c$: spontaneous magnetisation, $\langle M \rangle > 0$ without any external field
- $T > T_c$: $\langle M \rangle = 0$

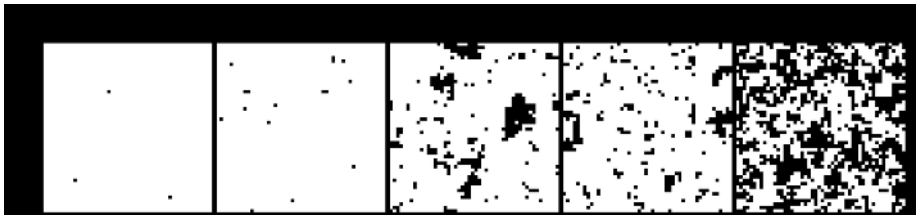


Monte Carlo averages for $L = 10, 20, 40$ (our data);
dashed red: Onsager's solution ($L \to \infty$); dotted: T_c

M is the **order parameter**: a quantity that tells the two phases apart. Near T_c: $M \sim (T_c - T)^{1/8}$, and fluctuations of observables become large.

Sampling Spin Configurations

$2^{L \cdot L}$ configurations are too many to list: Metropolis Monte Carlo draws samples from $p(\{s_i\}) \propto e^{-H/T}$ instead.

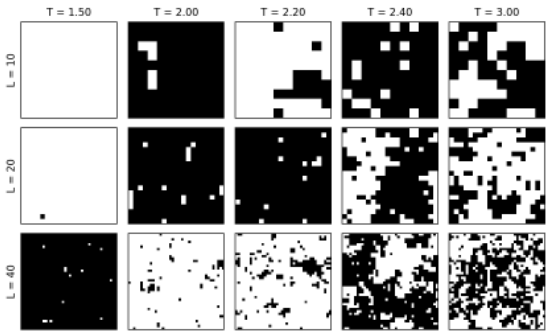


50×50 lattice at $T = 1.2,\ 1.9,\ 2.2,\ 2.4,\ 3.2$ (black and white: the two spin values)

Low T: spins aligned. High T: a random pattern.

N. Metropolis, A. W. Rosenbluth, M. N. Rosenbluth, A. H. Teller, E. Teller, *Equation of State Calculations by Fast Computing Machines*, J. Chem. Phys. **21**, 1087 (1953), doi:10.1063/1.1699114

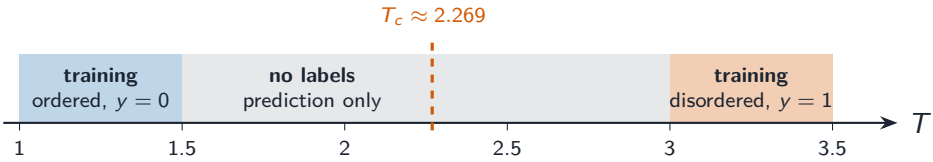
The Dataset



- Metropolis samples: 51 temperatures from 1.0 to 3.5, 200 configurations each, for $L = 10, 20, 40$
- each configuration is a black-and-white $L \times L$ image
- low T: almost all spins aligned; high T: random pattern
- on a small lattice $T = 2.2$ and $T = 2.4$ look alike, but lie on **opposite sides** of $T_c \approx 2.269$

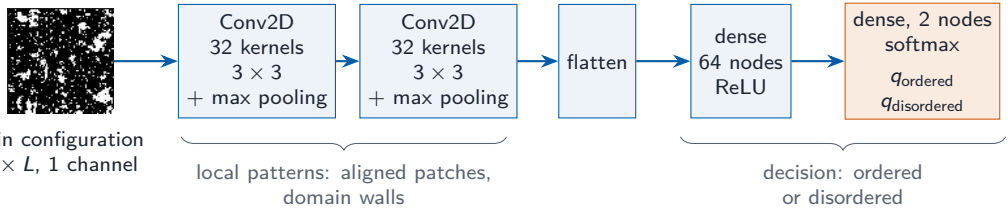
Task: find T_c from the images alone, without being told about the magnetisation.

Supervised Learning Far from the Transition



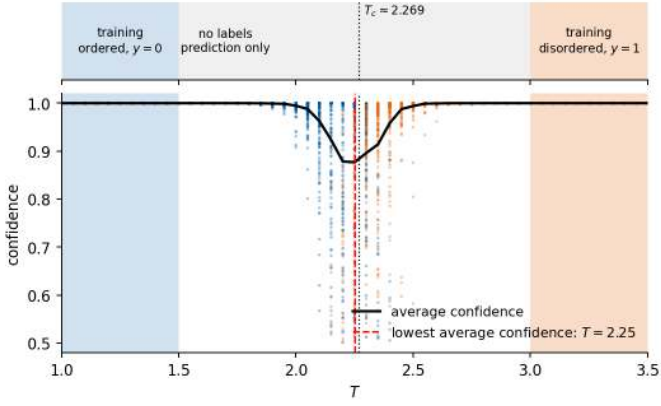
- **train only where the answer is clear:** $T < 1.5$ ordered, $T > 3$ disordered; configurations in between are left out
- **data:** input X = a spin configuration (an $L \times L$ image), label y = phase
- **predict** at every temperature, also where the network has seen no data, and plot the **confidence**: the softmax probability of the predicted class

The CNN for the Ising Model



- the same building blocks as for CIFAR-10: a spin configuration is a black-and-white image
- two outputs, softmax, categorical cross-entropy, Adam optimiser
- $q_{\text{ordered}} + q_{\text{disordered}} = 1$; the **confidence** is the larger of the two

The Confidence Drops at the Critical Temperature



$L = 40$; dots: the confidence of every configuration; line: its average. Lowest average confidence at $T = 2.25$; over three trainings 2.27 ± 0.02, exact $T_c \approx 2.269$.

Near T_c the magnetisation vanishes and fluctuations are large: the two phases are hardest to tell apart.

A First Discovery with a Neural Network

What we gained

- an estimate of T_c from raw configurations, with no formula for the order parameter
- for most physical systems the order parameter is *not known*: the network signals that *something happens* near a given temperature

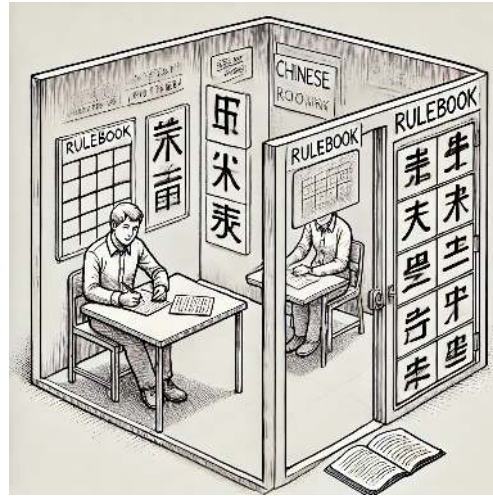
Limits

- finite lattice: the transition is smoothed, the dip is broadened; larger L gives a sharper estimate
- the labels and the two training windows came from us
- the network does not tell us *why* there is a transition

The network located the phase transition; the physics that explains it is still ours to find.

The Chinese Room Thought Experiment

- a person who does not understand Chinese sits inside a room
- a rulebook (an algorithm) says how to manipulate Chinese characters to answer questions written in Chinese
- to an observer outside, the room understands Chinese: the answers are coherent
- the person understands nothing; they only follow rules



Searle: manipulating symbols (syntax) is not enough for understanding (semantics).

J. R. Searle, *Minds, brains, and programs*, Behav. Brain Sci. **3**, 417 (1980), doi:10.1017/S0140525X00005756

The Chinese Room: Replies

Searle's paper lists replies to the argument and answers them.

- **systems reply:** the person does not understand, but the whole system (person, rulebook, paper) does.
  Searle: let the person memorise the rulebook; they still understand nothing.

- **robot reply:** connect the symbols to the world through cameras and motors, and they acquire meaning.
  Searle: the sensor data are just more symbols for the person in the room.

- **brain simulator reply:** simulate the neurons of a Chinese speaker instead of following a rulebook.
  Searle: simulating the formal structure of the brain still does not give its understanding.

The brain simulator reply is close to what a neural network is; the debate is still open.

J. R. Searle, *Minds, brains, and programs*, Behav. Brain Sci. **3**, 417 (1980), doi:10.1017/S0140525X00005756

The Chinese Room and Deep Learning

- **data manipulation:** a network maps inputs to outputs by learned transformations, as the person in the room follows rules
- **statistical, not semantic:** the network's knowledge is a set of correlations in the data; it does not know what the pixels, words or spins mean
- **subsymbolic:** the room uses explicit rules; a network uses weights and activations learned from data

Back to the Ising model

The CNN located T_c without an explanation: no order parameter, no mechanism of the transition. The understanding (Onsager's solution, the order parameter) remains the scientist's job.

Summary

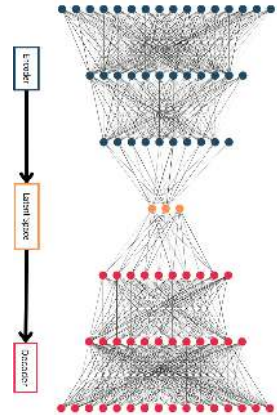
- **convolution**: small shared kernels see local patterns; CIFAR-10 test accuracy from 41% (dense network) to 88% (CNN with batch normalisation and augmentation)
- **regularisation**: L1/L2, dropout, batch normalisation, augmentation close the gap between training and test
- **class activation maps** show where the evidence for a class lies; Grad-CAM extends them to any CNN
- **adversarial examples**: a change invisible to us can flip the prediction
- **blood cells and satellite images**: models with similar accuracy make different mistakes; the confusion matrix and the class activation map show which
- **2D Ising model**: a CNN trained far from the transition locates T_c from spin configurations alone; explaining the transition is still our job (the Chinese room)

Next Lecture: Unsupervised Learning

Most scientific data come without labels: microscope archives, sky surveys, simulations, sequencing data.

Unsupervised learning finds structure in data without labels:

- **dimensionality reduction**: PCA, t-SNE, UMAP
- **clustering**: groups of similar samples, e.g. cell types
- **autoencoders**: compress the data to a few latent variables and reconstruct it
- **anomaly detection**: samples the model cannot reconstruct



Example: the phases of the Ising model found from spin configurations alone, with no temperature labels. L. Wang, *Discovering phase transitions with unsupervised learning*, Phys. Rev. B **94**, 195105 (2016), doi:10.1103/PhysRevB.94.195105

Thank You!

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